

Predicting Partition Coefficients from $\log P$ Using the Polarity Index P'

Synopsis

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Directly substituting partition coefficients or Henry-like constants (k) with $\log P$ values is  theoretically unsound, as no established framework supports such an approach.

This document presents a  generalized method based on Flory-Huggins theory , introducing a robust polarity index (P') . By deriving P' from $\log P$ and molecular mass (M) , this approach provides a  more consistent and predictive framework for estimating partition coefficients across different systems .

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Objective

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The goal is to establish a qualitative correlation between an approximate “polarity index” (P') and the octanol-water partition coefficient ($\log P$) for small organic molecules (i), using the Flory-Huggins (FH) theory of mixtures. The polarity index P' is introduced as an alternative to the square root of molar cohesive energies $\sqrt{E}$ in the approximation of FH interaction parameters for a solute i in a fluid or polymer phase k :

$$\chi_{i+k} \approx 0.34 \times (\sqrt{E_i} - \sqrt{E_k})^2 \quad (1)$$

where E_i and E_k are the cohesive energies of i and k , respectively. The empirical coefficient 0.34 is derived from experimental data.

References

- Guillaume Gillet, Olivier Vitrac, and Stéphane Desobry Prediction of Solute Partition Coefficients between Polyolefins and Alcohols Using a Generalized Flory–Huggins Approach *Industrial & Engineering Chemistry Research* 2009 48 (11), 5285-5301 <https://doi.org/10.1021/ie801141h>

- Thomas Brouwer and Boelo Schuur Model Performances Evaluated for Infinite Dilution Activity Coefficients Prediction at 298.15 K *Industrial & Engineering Chemistry Research* 2019 58 (20), 8903-8914 <https://doi.org/10.1021/acs.iecr.9b00727>

Derivation of the Polarity Index (P') from $\log P$

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Since $\log P$ values include a significant entropic contribution, particularly for large substances, this effect is removed using the same FH theory. For solute i , $\log P$ is related to the octanol-water partition coefficient $\ln K_{i,o/w}$ at infinite dilution:

$$\log P = \frac{\ln K_{i,o/w}}{\ln 10} \quad (2)$$

where:

$$\ln K_{i,o/w} = \gamma_{i,w} - \gamma_{i,o} \quad (3)$$

At infinite dilution in $k = o, w$, the FH theory gives the following expression for the activity coefficient of i in phase k :

$$\gamma_{i,k} \approx \chi_{i+k} + 1 - r_{i,k} \quad (4)$$

where $r_{i,k} \approx \frac{V_i}{V_k}$ is the molar volume ratio between i and k . As a result:

$$\ln 10 \times \log P \approx \Delta \chi_i^{scale:ow} + S_i^{correction:ow} \quad (5)$$

where the correction term accounts for entropic effects:

$$S_i^{correction:ow} = - (r_{i,w} - r_{i,o}) \quad (6)$$

By generalizing the FH coefficient approximation with a scaling parameter α and substituting cohesive energies with their corresponding polarity indices P' , we obtain:

$$\Delta \chi_i^{scale:ow} = \chi_{i,w} - \chi_{i,o} \approx \alpha \times [(P'_i - P'_w)^2 - (P'_i - P'_o)^2] \quad (7)$$

which expands to:

$$\begin{aligned} &= \alpha \times [P'_i \cdot (P'_w - P'_o) + (P'_w + P'_o)(P'_w - P'_o)] \\ &= \alpha \cdot (P'_w - P'_o) \times (P'_i + \beta) \\ &= A \cdot P'_i + B \end{aligned} \quad (8)$$

where A and B are constants.

Since P' is defined based on differences, we can set $B = 0$ without affecting predictions.

Definition and Range of P'

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- P' is designed to be a bounded measure of polarity, ranging from 0 (highly hydrophobic substances) to 10.2 (water, the most polar reference).
- It follows that:

$$A \cdot P' \approx \ln 10 \times \log P - S_i^{correction:ow}(V_i) \approx U(\log P, V_i) \quad (9)$$

where choosing $A = 1$ leads to:

$$\alpha = \frac{1}{(P'_w - P'_o)} \quad (10)$$

Empirical fitting suggests $\alpha \approx 0.14$. To ensure a smooth and bounded prediction of P' , we approximate it using a second-degree polynomial regression:

$$\hat{P}'(M, V) = a \cdot [U(\log P, V_i = f(M))]^2 + b \cdot U(\log P, V_i = f(M)) + c \quad (11)$$

where V_i is estimated using Miller's empirical formula:

$$V_i = f(M_i) \approx 0.997 \cdot M_i^{1.03} \quad (12)$$

with M_i in $g \cdot mol^{-1}$ and V_i in $cm^3 \cdot mol^{-1}$.

Fitting Methodology

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- The parameters a , b , and c are optimized based on reported $\log P$ and P' values from open databases.
- The upper bound of P' is determined by water's polarity, which depends on database values for $\log P$ and parameter choices.

Application to Activity Coefficients

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Once P' is derived, it can be used to estimate activity coefficients:

$$\begin{aligned} \gamma_{i,k} &\approx \chi_{i+k} + 1 - (r_{i,k} - n(r_{i,k})) \\ &\approx \alpha \times [\hat{P}'(\log P_i, V_i) - \hat{P}'(\log P_k, V_k)]^2 + 1 - (r_{i,k} - n(r_{i,k})) \end{aligned} \quad (13)$$

The correction term $n(r_{i,k})$ accounts for the presence of surrounding molecules within the effective volume of the solute. This effect is significant for large, flexible solutes and negligible for small, rigid molecules.

Special Cases

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- For polymers ($k = P$): Since the solute is considered infinitely small compared to the polymer, $r_{i,k} \rightarrow 0$ and $n(r_{i,k}) \rightarrow 0$.
- For food simulants ($k = F$): For a generalized approximation applicable to various substances, we propose:

$$n(r) = \frac{r - 5}{r} \quad (14)$$

References

Guillaume Gillet, Olivier Vitrac, and Stéphane Desobry

Prediction of Partition Coefficients of Plastic Additives between Packaging Materials and Food Simulants *Industrial & Engineering Chemistry Research* 2010 49 (16), 7263-7280 <https://doi.org/10.1021/ie9010595>

Summary

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- ✓ The methodology is theoretically well-founded and aligns with FH theory.
- ✓ It provides a practical way to estimate activity coefficients without requiring direct cohesive energy measurements.
- ✓ It accounts for entropy corrections, making P' a more transferable descriptor.
- ✓ The polynomial fit is simple and efficient, but its generalization across all chemical systems remains to be tested.
 - P' is introduced as a polarity index replacing cohesive energy terms in FH calculations.
 - It is directly linked to $\log P$ and fitted using second-degree polynomial regression.
 - The method provides a universal approach to predicting activity coefficients in various media, including polymers and food simulants.
 - A correction term $n(r)$ accounts for molecular interactions in complex environments.

Outlook

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- △ More validation is needed, especially for extreme cases (very high or low $\log P$) and polymers.
- △ Alternative functional forms for P' may improve accuracy.
- △ The empirical scaling factor (α) and correction function ($n(r)$) should be tested for robustness.



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